```
import matplotlib.pyplot as plt

plt.figure(figsize=(6,2))

plt.text(0.5,0.6,"Sowndarya Sree T",ha="center",fontsize=18)

plt.text(0.5,0.4,"727624BAM052",ha="center",fontsize=14)

plt.axis("off")
plt.show()
```

Sowndarya Sree T
727624BAM052

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("/content/ev_fleet_dataset_v15.csv")

df.head()
```

	vehicle_id	driver_id	driver_name	manufacturer	monthly_revenue_generation	yearly_revenue	maintenance_cost	driver_re
0	EV-103	103	Manikandan	Tesla	289059.48	3222512.77	24736.59	55
1	EV-436	436	Arun	Tata	399923.92	5127508.07	17823.20	36
2	EV-861	861	Vetri	Maruti	106210.59	1153897.98	12474.27	22
3	EV-271	271	Chandran	Mahindra	12673.89	138429.34	25721.08	35
4	EV-107	107	Suriya	BMW	154274.96	1662185.19	32126.37	23

Next steps: Generate code with df New interactive sheet

```
df.shape
```

```
(10000, 15)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 15 columns):
 #   Column                      Non-Null Count  Dtype  
---  ------                      --------------  -----  
 0   vehicle_id                  10000 non-null  object 
 1   driver_id                   10000 non-null  int64  
 2   driver_name                 10000 non-null  object 
 3   manufacturer                10000 non-null  object 
 4   monthly_revenue_generation  10000 non-null  float64
 5   yearly_revenue              10000 non-null  float64
 6   maintenance_cost            10000 non-null  float64
 7   driver_revenue              10000 non-null  float64
 8   battery_percentage          10000 non-null  int64  
 9   expected_distance_km        10000 non-null  float64
 10  trip_speed_kmph             10000 non-null  float64
 11  vehicle_status              10000 non-null  object 
 12  date                        10000 non-null  object 
 13  time                        10000 non-null  float64
 14  violations                  8000 non-null   object 
```

```
dtypes: float64(7), int64(2), object(6)
memory usage: 1.1+ MB
```

```
list(df)
```

```
['vehicle_id',
 'driver_id',
 'driver_name',
 'manufacturer',
 'monthly_revenue_generation',
 'yearly_revenue',
 'maintenance_cost',
 'driver_revenue',
 'battery_percentage',
 'expected_distance_km',
 'trip_speed_kmph',
 'vehicle_status',
 'date',
 'time',
 'violations']
```

```
df.describe()
```

	driver_id	monthly_revenue_generation	yearly_revenue	maintenance_cost	driver_revenue	battery_percentage	expec
count	10000.000000	10000.000000	1.000000e+04	10000.000000	10000.000000	10000.000000	
mean	504.470600	203879.814685	2.447606e+06	25659.699905	3133.839614	52.871600	
std	289.724434	126574.484429	1.528926e+06	9203.534781	1192.398937	27.911923	
min	1.000000	0.000000	0.000000e+00	4068.890000	200.000000	5.000000	
25%	251.000000	103955.352500	1.249759e+06	18625.995000	2266.350000	29.000000	
50%	506.500000	200442.515000	2.394791e+06	25377.485000	3066.735000	53.000000	
75%	758.000000	280523.895000	3.378185e+06	32070.227500	3936.722500	77.250000	
max	1000.000000	499796.540000	6.611489e+06	50000.000000	7478.170000	100.000000	

```
df.isnull().sum()
```

	0
vehicle_id	0
driver_id	0
driver_name	0
manufacturer	0
monthly_revenue_generation	0
yearly_revenue	0
maintenance_cost	0
driver_revenue	0
battery_percentage	0
expected_distance_km	0
trip_speed_kmph	0
vehicle_status	0
date	0
time	0
violations	2000

dtype: int64

```
running = df[df["vehicle_status"] == "Running"]

print("Vehicles Running =", len(running))
```

```
Vehicles Running = 7033
```

```
Charging = df[df["vehicle_status"] == "Charging"]

print("Vehicles Charging =", len(Charging))
```

```
Vehicles Charging = 1471
```

```
Garage = df[df["vehicle_status"] == "Garage"]

print("Vehicles Garage =", len(Garage))
```

```
Vehicles Garage = 1017
```

```
Maintenance = df[df["vehicle_status"] == "Maintenance"]

print("Vehicles Maintenance =", len(Maintenance))
```

```
Vehicles Maintenance = 479
```

```
count = df.groupby("manufacturer")["vehicle_id"].count()

print(count)
```

```
manufacturer
BMW         2000
Mahindra    2000
Maruti      2000
Tata        2000
Tesla       2000
Name: vehicle_id, dtype: int64
```

```
charging = df[df["battery_percentage"] < 20]

print("Vehicles Need Charging =", len(charging))
```

```
Vehicles Need Charging = 1535
```

```
cost = df.groupby("manufacturer")["maintenance_cost"].sum()

print(cost)
```

```
manufacturer
BMW         51257339.64
Mahindra    51377741.09
Maruti      51879992.78
Tata        50616071.45
Tesla       51465854.09
Name: maintenance_cost, dtype: float64
```

```
vehicle_cost = df.groupby("vehicle_id")["maintenance_cost"].sum()

print(vehicle_cost.head(20))
```

```
vehicle_id
EV-1       372919.53
EV-10      307755.64
EV-100     153532.38
EV-1000     96868.62
EV-101     320494.67
EV-102     286791.14
EV-103     207952.54
EV-104     316862.76
EV-105     206853.65
EV-106     292169.71
EV-107     224126.59
EV-108     220068.20
EV-109     422858.09
EV-11      142080.13
EV-110     165119.82
EV-111     420369.53
EV-112     370330.82
EV-113     452321.84
EV-114     266828.28
EV-115     143429.71
Name: maintenance_cost, dtype: float64
```

```
driver = df.groupby("driver_name")["driver_revenue"].sum()

print("Driver Name :", driver.idxmax())
print("Revenue :", f"{driver.max():,.0f}")
```

```
Driver Name : Boopathi
Revenue : 490,930
```

```
loss = df.groupby("driver_name")["maintenance_cost"].sum()
```

```
print("Driver Name :", loss.idxmax())
print("Loss :", f"{loss.max():,.0f}")
```

```
Driver Name : Vaishnavi
Loss : 4,020,363
```

```
v = df[df["violations"] != "None"]

result = v.groupby("driver_name").size()

print("Driver Name :", result.idxmax())
print("Violations :", result.max())
```

```
Driver Name : Vaishnavi
Violations : 154
```

```
df["date"] = pd.to_datetime(df["date"], dayfirst=True)

df["month"] = df["date"].dt.month_name()

month_revenue = df.groupby("month")["monthly_revenue_generation"].sum()

for month, revenue in month_revenue.items():
    print(month, ":", f"{revenue:,.0f}")
```

```
April : 161,519,534
August : 170,310,399
December : 174,537,919
February : 155,702,195
January : 169,483,106
July : 172,550,766
June : 168,234,548
March : 172,028,334
May : 178,335,084
November : 175,894,544
October : 177,758,712
September : 162,443,005
```